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Re: AI Action Plan

Mr. D'Souza:

Merlin Labs, Inc. ("Merlin") respectfully submits these comments in response to the RFI on the Development of an Artificial Intelligence (AI) Action Plan ("Plan"), published on February 6, 2025 by the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), on behalf of the Office of Science and Technology Policy (OSTP).

We are pleased that the OSTP is developing policies with the aim to further enable American AI innovation. At Merlin, we believe AI technologies will transform society with deep implications for operational safety and global competitiveness of American aviation and defense. As a leading developer of advanced flight technology, Merlin responds to this RFI with the distinct perspective of an aviation industry innovator in the civilian and defense sectors.

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the government in developing the AI Action Plan and associated documents without attribution.

ABOUT MERLIN

Merlin is building the Merlin Pilot, an advanced aviation system that will improve the safety and efficiency of flight operations by augmenting existing pilot functions and reducing overall workload — without requiring any updates to the existing infrastructure. The company was founded in 2018 and is headquartered in Boston, MA, with additional offices in Denver, CO and flight test facilities in Quonset, RI, and Kerikeri, New Zealand. Merlin is institutionally-backed from leading private sector venture capital firms, including First Round Capital, Google Ventures, and Ballie Gifford.

Merlin has successfully designed, integrated, and tested its technology on a variety of aircraft platforms, ranging from single engine piston aircraft to twin engine turboprops. Human-machine teaming is at the forefront of Merlin's technology development process as the company believes a thorough consideration of human factors will ultimately enable the integration of highly automated systems on aircraft. Merlin is currently in the process of certifying its technology via an Supplemental Type Certificate on a Part 23 aircraft, and is actively collaborating with the US military to integrate its platform into transport category platforms.

OPPORTUNITIES AND CHALLENGES FOR AI IN AVIATION

The key driver for Merlin's technology is improved safety in aviation operations across a variety of applications. With regards to artificial intelligence (AI), Merlin focuses on how to develop and deploy the technology in a manner that is commensurate with the high safety expectations of civilian and military aviation technologies that exist today. AI provides an opportunity to advance functionality on aircraft, though this must be done in a manner that is safe, reliable, and demonstrable.

Existing standards, policy, and guidance provide an effective infrastructure that can be leveraged to support applications in this space, but gaps in this infrastructure do exist and must not be ignored. Merlin has been working with industry committees and government organizations, domestically and internationally, to assess these gaps and find pragmatic solutions to close them.

The key challenge in the developments in AI safety assurance is understanding the scope of the problem and safety implications. Governments around the world have

differing views on the extent of the concerns regarding AI and how to address them. For Merlin, the structure that current aviation regulations and policy provide makes this problem clear.

Advanced automation is not new to aviation. The industry, in coordination with regulators, has worked to develop standards that explain and address how these complex functions can be shown to be safe. These standards involve development assurance frameworks that systematically decompose a highly complex and integrated aircraft into smaller, more comprehensible systems, and those systems into bounded software. These assurance methods provide transformational integrity that the divided problem of the aircraft can be developed in these smaller elements and also be brought together to show they do all that is expected and only what is acceptable.

These frameworks, as defined in standards like SAE ARP4761A, SAE ARP4754B, RTCA DO-178C, and DO-254, help developers understand that AI can only be implemented in elements already bound by these processes. In fact, in order to rely on conventional protections, such as runtime assurance, to guard against unsafe behavior by AI, a developer has to prove conventionally that those protections are capable of having their intended effect when the AI acts unsafely. To show these conventional protections work, we use conventional assurance processes. It is not simply that we *can* rely on these existing processes and standards; it is that we *must* rely on them.

PRIORITY ITEMS TO INCLUDE IN THE AI ACTION PLAN

As AI continues to transform the aviation industry, a clear and collaborative policy framework is essential to ensure safe and effective integration. Merlin recommends that the OSTP consider the following items in its AI Action Plan to support innovation while maintaining safety and regulatory clarity:

- 1. Streamline a Pathway for Integrating AI Technologies in Aviation**
- 2. Continue Efforts the STEP Program to Facilitate Industry Discussions on AI**
- 3. Set Clear Regulatory Thresholds for AI Systems Based on Safety Impact**
- 4. Create Aviation Communication Datasets for Algorithm Training**

Item 1: A streamlined pathway for integrating AI in aviation

The AI Action Plan should support policy approaches that enable a streamlined pathway for integrating AI in aviation. Current regulations require developers using AI on an aircraft system to undergo the Issue Paper process, regardless of the application's safety criticality. While Issue Papers are a well-established tool for certification, applying this requirement to all AI applications creates an unnecessary burden, particularly for low-risk, non-safety critical functions. By definition, these applications do not cause or contribute to safety critical failure conditions and should not require additional safety and compliance measures. This requirement discourages applicants from using AI in low-criticality contexts and inhibits the industry from gathering needed experience with the technology.

This challenge is especially evident in the context of Non-Required Safety Enhancing Equipment (NORSEE), a framework designed to streamline the certification of safety-enhancing technologies. NORSEE has successfully facilitated the integration of tools like electronic flight bags (EFBs) to improve pilot situational awareness. However, AI-based NORSEE technologies are still subject to the Issue Paper process. To address this undue burden, guidance should exempt non-safety critical AI applications from additional compliance requirements and expand NORSEE's applicability to AI-driven innovations across Part 25 operations. This adjustment would accelerate the introduction of beneficial technologies while maintaining aviation's high safety standards.

Item 2: Continuing efforts by the FAA STEP Program to facilitate industry discussions

The AI Action Plan should support ongoing and future efforts for industry to collaborate on and discuss with regulators on policy development related to AI. In recent years, the FAA has played a key role in supporting the view that existing standards and processes provide a useful framework for demonstrating the safety of AI-driven implementations. The FAA AI Summits and meetings, run by the FAA STEP Program, have fostered industry coordination around this framework and have enabled the industry to see the advantages of this approach. Further, the FAA supports an experienced-based approach to develop new policy and guidance, which Merlin believes will result in more pragmatic solutions to the eventual demonstration of the safety for AI.

Merlin encourages the OSTP to include policy language in its AI Action Plan that would support the continued efforts of the STEP program and similar opportunities to

facilitate important industry discussions on AI best practices. Such meetings counter the voices of those who would make the challenges more complex by ignoring the mature infrastructure which can already solve the problems they are identifying.

Item 3: Setting clear regulatory thresholds based on safety impacts

The AI Action Plan should encourage the FAA to apply the aviation industry's established safety continuum framework and functional safety principles to set clear, risk-based thresholds for AI systems.

Specifically:

- Low safety impact AI applications (e.g., those with an Item Development Assurance Level [IDAL] of D or lower) should be subject only to existing regulatory frameworks, without additional certification burdens.
- Higher safety impact applications should undergo proportionally increased scrutiny, following established aviation certification principles.

To implement this, The AI Action Plan should support guidance development that:

- Prevents unnecessary regulatory barriers for low-risk AI innovations.
- Ensures appropriate oversight where AI poses genuine safety concerns.
- Aligns AI certification pathways with existing safety frameworks.

Current guidance is structured in such a way that low safety critical applications do not have to reveal how they are coded nor how well the algorithms perform. It is an excess burden to put higher expectations on a particular software development only because the regulator is aware of how it will be developed. Merlin recommends that any additional objectives a regulator may propose for low safety criticality items be considered not necessary for approval, but beneficial for future research in approval demonstration.

Item 4: Creating aviation communications datasets for NLP training

Lastly, the AI Action Plan should make available resources that enable developers to access the data they crucially need to build robust and safe AI-driven systems. For example, developing Natural Language Processing (NLP) algorithms for aviation

communications systems requires large datasets of existing communications that can be used to train and verify models. These datasets must capture the breadth and variety of today's aviation communications, and creating them is inherently a collaborative effort between regulators, industry, and academia to ensure that results are comprehensive, sufficient, and impartial.

We believe the AI Action Plan should implement policies that would enable the FAA to support such efforts, such as by launching an initiative to create publicly available datasets for the purposes of advanced technology development. Such an effort would be similar to how the FAA previously worked collaboratively with MIT Lincoln Laboratory and broader industry to develop airspace encounter models, which subsequently supported standards creation and development of collision avoidance systems

Establishing an effort to create training datasets for aviation communications would immediately accelerate the development of systems that enhance safety within the NAS, as well as inform standards and other policymaking efforts to formalize verification and validation of NLP algorithms going forward.